



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	COMMUNICATION ELECTRONICS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 – 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY2CJ101- ELECTRONICS I & PHY6CJ305- ELECTRONICS II				
Course Summary	Communication Electronics delves into the theory and practical implementation of electronic circuits and systems used in telecommunications, covering topics such as modulation techniques, signal processing, and transmission line theory, to facilitate efficient and reliable communication networks.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate an understanding of amplitude and frequency modulation techniques and their applications	Understand	Conceptual Understanding	Examinations, Assignments
CO2	Apply pulse and digital modulation techniques to design and analyze communication systems	Apply	Application	Problem Sets, Lab Reports
CO3	Analyze the components and operation of radio transmitters, receivers, and antennas in communication systems	Analyze	Application	Research Papers, Projects
CO4	Evaluate the principles and techniques of digital signal processing as applied to communication systems	Evaluate	Application	Presentations, Discussions
CO5	Explain the functionality and characteristics of radio transmitters, receivers, and antennas	Understand	Conceptual Understanding	Written Reports, Essays
CO6	Synthesize knowledge of modulation techniques, radio systems, and digital signal processing for designing and implementing communication systems	Create	Synthesis	Capstone Projects, Oral Defenses
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	AMPLITUDE AND FREQUENCY MODULATION TECHNIQUES		12	15
	1	Theory of Amplitude modulation	2	
	2	Double side band suppressed carrier Technique (DSBSC) and Single side band technique (SSB)	3	

	3	Generation of AM, DSBSC and SSB signals.	1	
	4	Theory of Angle modulation, Frequency modulation, Phase modulation and comparison of Frequency and Phase modulation	2	
	5	Frequency spectrum of FM wave, Narrow band and wide band FM, Noise and Frequency modulation	2	
	6	Pre-emphasis and De-emphasis, Comparison of AM and FM	2	
	Relevant sections from Book 1			
II	PULSE AND DIGITAL MODULATION TECHNIQUES		10	15
	7	Pulse Amplitude Modulation, Pulse width Modulation, Pulse Position Modulation	4	
	8	Demodulation of Pulse modulated signals	3	
	9	Amplitude Shift keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK)	3	
	Relevant sections from Book 1			
III	RADIO TRANSMITTERS, RECEIVERS AND ANTENNAS		14	25
	10	AM, FM and SSB transmitters	1	
	11	Tuned Radio frequency receiver (TRF), Superheterodyne receiver, AM receiver	1	
	12	RF section and characteristics, frequency changing and tracking, ,	1	
	13	Intermediate frequencies and IF amplifiers, Automatic Gain control	1	
	14	FM Receiver and Ratio detector	1	
	15	Antennas, Potential functions and the EM field	2	
	16	Radiation from an oscillating dipole, far field and near field approximations	2	

	17	Power radiated by a current element, Radiation resistance of a short dipole, Radiation from a quarter wave monopole (qualitative ideas only)	2	
	18	Directivity – Gain and effective aperture of an antenna	1	
	19	Antenna arrays – Two element, linear and binomial, Frequency independent antennae, Log periodic antennae, Yagi antennae.	2	
	Relevant sections from Boos 1, 2 and 3			
IV	ELEMENTS OF DIGITAL SIGNAL PROCESSING TECHNIQUES		12	15
	20	Classifications of signals, concept of frequency in continuous-time and discrete-time signals	2	
	21	Theory of A/D and D/A conversion, Sampling of Analog signals, sampling Theorem	3	
	22	Quantization of continuous amplitude signal, Coding of quantized samples, Discrete time, linear time invariant systems	2	
	23	Techniques of analysis of linear systems, Resolution of a discrete time signal into impulses, Response of LTI systems to arbitrary inputs	2	
	24	Convolution sum - properties of convolution and the interconnection of LTI systems, Casual LTI systems, Stability of LTI systems.	3	
	Relevant sections from Books 4			
V	OPEN ENDED MODULE		12	

1	<u>Elements of communication system</u>: Need for modulation, Basics of signal representation and analysis, sine wave and Fourier series review, Frequency spectra of Non-sinusoidal Waves, Noises in signals, Signal-to-noise ratio, <u>Broadband communication systems</u>:- Multiplexing, frequency and time division multiplexing, Short and Medium Haul systems, Coaxial cables, fiber-optic links, Microwave links <u>Propagation of radio waves</u> - Ground waves, Sky wave propagation, Space waves, Tropospheric scatter propagation, Extra terrestrial communication. Ionosphere – Reflection and refraction of waves by the ionosphere – Attenuation,	12	
	Relevant sections from Books 1 and 5		

Books and References:

1. Electronic Communication Systems, 5th Edition, George Kennedy, B. Davis, S. R. M Prasanna, McGraw Hill, 2015 (Book 1)
2. Fundamentals of Applied Electromagnetics, Fawwaz T Ulaby, Pearson Education (Book 2)
3. Electromagnetic waves and Radiating Systems, Jordan E. C. and Balmain, K. G. , Prentice Hall India Ltd. (Book 3)
4. Digital Signal Processing, Proakis and Manolakis, Prentice Hall of India (1997) (Book 4)
5. Electronic Communications, Dennis Roddy and John Coolen, J., Pearson Education, Dorling Kindersley (India) Pvt. Ltd. (Book 5)
6. Foundations of Antenna Theory and Techniques, Vincent F. Fusco, Pearson Education Limited
7. Fundamentals of Communication Systems, John G Proakis, Masoud Salehi, Pearson Education
8. Antenna and Wave Propagation, John D Kraus
9. Digital Signal Processing, C. Ramesh Babu Durai, Laxmi Publications, New Delhi
10. An Integrated Course in Electronics & Communication Engineering, J B Gupta, S.K. Kataria & Sons Educational Publisher
11. Signals and Systems, Alan V. Oppenheim and Alan S. Willsky, Prentice Hall series

Mapping of COs with PSOs and POs :

	PSO 1	PS O2	PS O3	PSO 4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	0	0	0	0	2	2	2	0	2	0	0
CO 2	3	3	0	0	0	0	2	2	2	0	2	0	0
CO 3	3	3	3	0	0	0	2	2	2	0	2	0	0
CO 4	3	3	3	2	0	0	2	2	2	0	2	0	0
CO 5	3	2	0	0	0	0	2	2	2	0	2	0	0
CO 6	3	2	2	2	0	0	2	2	2	0	2	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓